Early-Growth Thresholds for the SIS Attendance Model

Setup. Let A be the symmetric adjacency matrix of an undirected contact graph. For attendance vector $d_t \in \{0,1\}^n$, set $D_t = \mathrm{diag}(d_t)$ and $B_t = D_t A D_t$. Let $X_i(t)$ indicate infection at the start of day t, let $p_i(t) = \Pr(X_i(t) = 1)$, and put $a = 1 - \gamma$, where $0 < \gamma < 1$. We use the rare-infection approximation in which probabilities of simultaneous infections of two specified individuals are second-order terms.

Proposition 1 (Linearized 14-day growth)**.** *To first order in the infection probabilities,*

$$p_{t+1} = M_t(\beta) p_t + O(\|p_t\|^2), \qquad M_t(\beta) = aI + \beta B_t. \tag{1}$$

For a 14-day attendance cycle, define

$$P_{14}(\beta) = M_{13}(\beta) M_{12}(\beta) \cdots M_0(\beta), \qquad g_{14}(\beta) = \rho(P_{14}(\beta)). \tag{2}$$

The linearized threshold β_c satisfies $g_{14}(\beta_c) = 1$.

Proof. Conditional on the start-of-day state, the SIS rule gives

$$\mathbb{E}[X_i(t+1) \mid X(t)] = aX_i(t) + (1 - X_i(t)) \left[1 - (1-\beta)^{\sum_j (B_t)_{ij} X_j(t)}\right]. \tag{3}$$

With zero infected neighbors the bracket is zero, and with one it equals β. Events with at least two infected neighbors and products $X_i(t)X_j(t)$ contribute second-order terms under the stated approximation. Taking expectations therefore yields

$$p_i(t+1) = a p_i(t) + \beta \sum_j (B_t)_{ij} p_j(t) + O(\|p_t\|^2),$$

which is equation (1). Repeated substitution gives $p_1 \simeq M_0 p_0$, $p_2 \simeq M_1 M_0 p_0$, and $p_{14} \simeq P_{14} p_0$ with the order in equation (2). The dominant per-cycle multiplier is the spectral radius $\rho(P_{14})$. □

Proposition 2 (Schedule-specific thresholds)**.** *Let $\lambda_F = \lambda_{\max}(A)$. Let A_{00} and A_{11} be the adjacency matrices induced by the two rotating groups, and define*

$$\lambda_R = \max\{\lambda_{\max}(A_{00}), \lambda_{\max}(A_{11})\}.$$

Let A_H be the graph obtained by permanently retaining a random half of the original edges in the On-Off control, and put $\lambda_H = \lambda_{\max}(A_H)$. For positive relevant eigenvalues,

$$\beta_c^{\mathrm{Full}} = \frac{\gamma}{\lambda_F}, \tag{4}$$

$$\beta_c^{\mathrm{OnOff}} = \frac{a^{-1} - a}{\lambda_F}, \tag{5}$$

$$\beta_c^{\mathrm{Rotating}} = \frac{a^{-1} - a}{\lambda_R}, \tag{6}$$

$$\beta_c^{\mathrm{HalfEdge}} = \frac{a^{-1} - a}{\lambda_H}. \tag{7}$$

The daily and weekly versions of each 50%-attendance schedule have the same linearized 14-day threshold. If a relevant eigenvalue is zero, its corresponding threshold is $+\infty$.

Proof. Because A is symmetric and entrywise nonnegative, the leading eigenvalue of $aI + \beta A$ is $a + \beta\lambda_F$ for $\beta \geq 0$. Under Full attendance,

$$g_{14}^{\text{Full}}(\beta) = \rho\big((aI + \beta A)^{14}\big) = (a + \beta\lambda_F)^{14}.$$

Equating this expression to one gives $a + \beta_c\lambda_F = 1$, hence $\beta_c = (1 - a)/\lambda_F = \gamma/\lambda_F$.

For On-Off, any complete 14-day cycle has seven on days with matrix $aI + \beta A$ and seven off days with matrix aI. The off-day matrix commutes with the on-day matrix, so either switching frequency has

$$P_{14}^{\text{OnOff}}(\beta) = (aI)^7(aI + \beta A)^7, \qquad g_{14}^{\text{OnOff}}(\beta) = \big[a(a + \beta\lambda_F)\big]^7.$$

Thus $a(a + \beta_c\lambda_F) = 1$, or $\beta_c = (a^{-1} - a)/\lambda_F$.

For Rotating, reorder vertices by group. The two daily matrices are

$$M^{(0)} = \begin{pmatrix} aI_0 + \beta A_{00} & 0 \\ 0 & aI_1 \end{pmatrix}, \qquad M^{(1)} = \begin{pmatrix} aI_0 & 0 \\ 0 & aI_1 + \beta A_{11} \end{pmatrix}.$$

They commute because they act on disjoint blocks. Each group attends seven days and stays home seven days, giving

$$P_{14}^{\text{Rotating}}(\beta) = \begin{pmatrix} a^7(aI_0 + \beta A_{00})^7 & 0 \\ 0 & a^7(aI_1 + \beta A_{11})^7 \end{pmatrix}.$$

Therefore $g_{14}^{\text{Rotating}}(\beta) = \big[a(a + \beta\lambda_R)\big]^7$, and equation (6) follows. Finally, the half-edge control has the On-Off attendance matrices with A replaced by A_H, which yields equation (7). □

Two-vertex check. For one edge connecting two vertices, $\lambda_F = 1$. With $\gamma = 0.2$, $a = 0.8$ and $a^{-1} - a = 1.25 - 0.8 = 0.45$. Equation (5) gives $\beta_c^{\text{OnOff}} = 0.45$.